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Republic of the Philippines
CAVITE STATE UNIVERSITY

Bacoor City Campus
SHIV, Molino VI, City of Bacoor



Module 5: PHP and MySQL



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DEPARTMENT OF COMPUTER STUDIES



Republic of the Philippines
**CAVITE STATE
UNIVERSITY**
Bacoor City Campus
SHIV, Molino VI, City of Bacoor

MISSION

Cavite State University shall provide excellent, equitable and relevant educational opportunities in the arts, science and technology through quality instruction and relevant research and development activities. It shall produce professional, skilled and morally upright individuals for global competitiveness.

VISION

The premier university in historic Cavite recognized for excellence in the development of globally competitive and morally upright individuals.

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COURSE DESCRIPTION

ITEC 106: IT ELECTIVE 2 (WEB SYSTEMS AND TECHNOLOGIES 2)

The course covers deployment and serving media contents within web applications. Implementation of websites and integration with other applications is included. Students must also look at possible improvements in the implementation of web applications to enhance security and avoid vulnerabilities.

PROGRAM OUTCOMES ADDRESSED BY THE COURSE. AFTER COMPLETING THIS COURSE, THE STUDENTS MUST BE ABLE TO:

1. Attain the vision, mission, goals and objectives of the university, campus and department,
2. Deliver a gender fair and gender sensitive instruction to students aligned with University goals and objectives,
3. Explain different components and technologies of World Wide Web as a platform.
4. Design and develop websites using fundamental web languages, technologies, and tools.
5. Distinguish between server-side and client-side web technologies.
6. Describe various web technology and application development issues and trends.
7. Develop a web application related to the course material.

COURSE REQUIREMENTS

1. Major Examination
2. Student Portfolio / Activities
3. Class Participation



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INTENDED LEARNING OUTCOMES

After the completion of the unit, students will be able to:

1. Differentiate the functions used in connecting a database with PHP language.
2. Create a PHP program that supports connection with database.

PRE-TEST:

Direction: Identify the term describe by the following:

1. It is one of the most popular relational database used in the web today with PHP language.
2. This extension is more portable and supports more than twelve different databases.
3. This extension support MySQL database only.
4. This function in MySQLi allows the program to be connected in a MySQL Server.
5. This parameter specifies the host name or IP address of the MySQL server.
6. This function closes the connection to MySQL server.
7. What is the default username in MySQL database server?
8. This function executes the specified SQL query.
9. A statement in SQL that retrieves records from the database table.
10. A statement in SQL that modifies the record in a database table.

INTRODUCTION

- ☐ MySQL is the most popular database system used with the PHP language.

What is MySQL?

MySQL is one of the most popular relational database system being used on the Web today. It is freely available and easy to install, however if you have installed

Wampserver it already there on your machine. MySQL database server offers several advantages:

- ❑ MySQL is easy to use, yet extremely powerful, fast, secure, and scalable.
- ❑ MySQL runs on a wide range of operating systems, including UNIX or Linux, Microsoft Windows, Apple Mac OS X, and others.
- ❑ MySQL supports standard SQL (Structured Query Language).
- ❑ MySQL is ideal database solution for both small and large applications.
- ❑ MySQL is developed, and distributed by Oracle Corporation.
- ❑ MySQL includes data security layers that protect sensitive data from intruders.

MySQL database stores data into tables like other relational database. A table is a collection of related data, and it is divided into rows and columns.

CONNECT TO MYSQL SERVER

Ways of Connecting to MySQL through PHP

In order to store or access the data inside a MySQL database, you first need to connect to the MySQL database server. PHP offers two different ways to connect to MySQL server: MySQLi (Improved MySQL) and PDO (PHP Data Objects) extensions.

While the PDO extension is more portable and supports more than twelve different databases, MySQLi extension as the name suggests supports MySQL database only. MySQLi extension however provides an easier way to connect to, and execute queries on, a MySQL database server. Both PDO and MySQLi offer an object-oriented API, but MySQLi also offers a procedural API which is relatively easy for beginners to understand.

Connecting to MySQL Database Server

In PHP you can easily do this using the **mysqli_connect()** function. All communication between PHP and the MySQL database server takes place through this

connection. Here're the basic syntaxes for connecting to MySQL using MySQLi and PDO extensions:

The **hostname** parameter in the above syntax specify the host name (e.g. localhost), or IP address of the MySQL server, whereas the username and password parameters specifies the credentials to access MySQL server, and the database parameter, if provided will specify the default MySQL database to be used when performing queries.

Syntax: MySQLi, Procedural way

```
$var = mysqli_connect("hostname","username","password","database");
```

Example:

```
<?php
/* Attempt MySQL server connection. Assuming you are running MySQL
server with default setting (user 'root' with no password) */
$con = mysqli_connect("localhost", "root", "", "sampledb");

// Check connection
if($con === false){
    die("ERROR: Could not connect. " . mysqli_connect_error());
}

// Print host information
echo "Connect Successfully. Host info: " . mysqli_get_host_info($con);
?>
```

Syntax: MySQLi, Object Oriented way

```
$var = new mysqli("hostname","username","password","database");
```

Example:

```
<?php
/* Attempt MySQL server connection. Assuming you are running MySQL
server with default setting (user 'root' with no password) */
$con = new mysqli("localhost", "root", "", "sampledb");

// Check connection
if($con === false){
    die("ERROR: Could not connect. " . $con->connect_error);
}

// Print host information
echo "Connect Successfully. Host info: " . $con->host_info;
?>
```

Syntax: PHP Data Objects (PDO) way

```
$var = new PDO("mysql:host=hostname;dbname=database", "username",
"password");
```

Example:

```
<?php
/* Attempt MySQL server connection. Assuming you are running MySQL
server with default setting (user 'root' with no password) */
try{
    $con = new PDO("mysql:host=localhost;dbname=sampledb",
    "root","");

    // Set the PDO error mode to exception
    $con->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

    // Print host information
    echo "Connect Successfully. Host info: " .
    $con->getAttribute(constant("PDO::ATTR_CONNECTION_STATUS"));
}
catch(PDOException $e){
    die("ERROR: Could not connect. " . $e->getMessage());
}
?>
```

The default username for MySQL database server is root and there is no password. However to prevent your databases from intrusion and unauthorized access you should set password for MySQL accounts.

Setting the PDO::ATTR_ERRMODE attribute to PDO::ERRMODE_EXCEPTION tells PDO to throw exceptions whenever a database error occurs.

Closing the MySQL Database Server Connection

The connection to the MySQL database server will be closed automatically as soon as the execution of the script ends. However, if you want to close it earlier you can do this by simply calling the PHP **mysqli_close()** function.

Example: MySQLi, Procedural way

```
<?php  
    mysqli_close($con);  
?>
```

Example: MySQLi, Object Oriented way

```
<?php  
    $con->close();  
?>
```

Example: PHP Data Objects (PDO) way

```
<?php  
    unset($con);  
?>
```

CRUD APPLICATION IN PHP

The **mysqli_query()** function performs a query against a database.

Syntax: **mysqli_query(connection, query)**

The SQL **SELECT** statement is used to select the records from database tables. Create an SQL query using the SELECT statement, after that execute the SQL query by passing it to the PHP **mysqli_query()** function to retrieve the table data.

Example:

```
<table>
  <tr>
    <th>Customer ID</th>
    <th>Customer Name</th>
    <th>Country</th>
    <th>Contact No</th>
  </tr>
  <?php
    // Include database config file
    require_once "config.php";
    $sql="SELECT * FROM customer";
    $result=mysqli_query($con,$sql);
    while($row=mysqli_fetch_array($result)):?>
      <tr>
        <td><?php echo $row['CustomerID']; ?></td>
        <td><?php echo $row['CustomerName']; ?></td>
        <td><?php echo $row['Country']; ?></td>
        <td><?php echo $row['ContactNo']; ?></td>
        <td><a href="codesimulation.php?delete=<?php echo
$row['CustomerID'];?>">Delete</a></td>
      </tr>
    <?php endwhile;?>
</table>
```

The **INSERT INTO** statement is used to insert new rows in a database table. Create an SQL query using the INSERT INTO statement with appropriate values, after that execute the insert query by passing it to the PHP **mysqli_query()** function to insert data in table.

Example:

```
<?php
  // Include database config file
  require_once "config.php";
  if (isset($_POST['insert'])) {
    $cid=$_POST['cid'];
    $name=$_POST['name'];
    $country=$_POST['country'];
    $contact=$_POST['contact'];
    $sql="INSERT INTO customer
VALUES($cid,$name,$country,$contact)";
    if(mysqli_query($con,$sql)){
```



```

        echo "Record inserted successfully";
    }
    else
        echo "ERROR: Could not able to execute $sql. " .
mysqli_error($con);
    }
?>
<form method="POST">
    <input type="text" name="cid" placeholder="Customer ID">
    <input type="text" name="name" placeholder="Customer Name">
    <input type="text" name="country" placeholder="Country">
    <input type="text" name="contact" placeholder="Contact">
    <input type="submit" name="insert" value="Submit">
</form>

```

The **UPDATE** statement is used to change or modify the existing records in a database table. This statement is typically used in conjugation with the WHERE clause to apply the changes to only those records that matches specific criteria. Execute the query using PHP mysqli_query() function to update data in table.

Example:

```

<?php
    // Include database config file
    require_once "config.php";
    if (isset($_POST['update'])) {
        $cid=$_POST['cid'];
        $name=$_POST['name'];
        $country=$_POST['country'];
        $contact=$_POST['contact'];
        $sql="UPDATE customer SET CustomerName='$name',
Country='$country', ContactNo=$contact WHERE CustomerID=$cid";
        if(mysqli_query($con,$sql)){
            echo "Record updated successfully";
        }
        else
            echo "ERROR: Could not able to execute $sql. " .
mysqli_error($con);
    }
?>
<form method="POST">
    <input type="text" name="cid" placeholder="Customer ID">
    <input type="text" name="name" placeholder="Customer Name">
    <input type="text" name="country" placeholder="Country">
    <input type="text" name="contact" placeholder="Contact">
    <input type="submit" name="update" value="Submit">
</form>

```

Just as you insert records into tables, you can delete records from a table using the SQL **DELETE** statement. It is typically used in conjugation with the WHERE clause to delete only those records that matches specific criteria or condition. Execute the query using PHP **mysqli_query()** function to delete data in table.

Example:

```
<?php
    // Include database config file
    require_once "config.php";
    if (isset($_GET['delete'])) {
        $cid=$_GET['delete'];
        $sql="DELETE FROM customer WHERE CustomerID=$cid";
        if(mysqli_query($con,$sql)){
            echo "Record deleted successfully";
        }
        else
            echo "ERROR: Could not able to execute $sql. " .
mysqli_error($con);
    }
?>
<table>
    <tr>
        <th>Customer ID</th>
        <th>Customer Name</th>
        <th>Country</th>
        <th>Contact No</th>
    </tr>
    <?php
        // Include database config file
        require_once "config.php";
        $sql="SELECT * FROM customer";
        $result=mysqli_query($con,$sql);
        while($row=mysqli_fetch_array($result)):?>
            <tr>
                <td><?php echo $row['CustomerID']; ?></td>
                <td><?php echo $row['CustomerName']; ?></td>
                <td><?php echo $row['Country']; ?></td>
                <td><?php echo $row['ContactNo']; ?></td>
                <td><a href="codesimulation.php?delete=<?php echo
$row['CustomerID']; ?>">Delete</a></td>
            </tr>
        <?php endwhile;?>
    </table>
```

POST-TEST:

Direction: Identify the terms describe by the following:

1. It is one of the most popular relational database used in the web today with PHP language.
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3. This extension support MySQL database only.
4. This function in MySQLi allows the program to be connected in a MySQL Server.
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10. A statement in SQL that modifies the record in a database table.

PRE-TEST AND POST-TEST FEEDBACK

PRE-TEST

1. MySQL
2. PDO
3. MySQLi
4. mysqli_connect()
5. hostname
6. mysqli_close()
7. root
8. mysqli_query()
9. SELECT
10. UPDATE

POST-TEST

1. MySQL
2. PDO
3. MySQLi
4. mysqli_connect()
5. hostname
6. mysqli_close()
7. root
8. mysqli_query()
9. SELECT
10. UPDATE

REFERENCES

- Harris, A. (2011). *HTML, XHTML, & CSS All-in-One For Dummies, 2nd Edition*. Indianapolis, Indiana: Wiley Publishing, Inc.
- Noble, J. and Tittel, E. (2011). *HTML, XHTML & CSS FOR DUMMIES 7th Edition*. Indianapolis, Indiana: Wiley Publishing, Inc.
- W3schools (2020). HTML, CSS, Javascript and PHP. Retrieved from <https://www.w3schools.com/default.asp>
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